

# HOUSE PRICE PREDICTION SUMMARY

## TASK-1: Data Loading & Exploration

- The dataset contains 545 rows and 13 columns.
- The target variable is price.
- The remaining 12 columns are features used to predict house prices.
- Both numerical and categorical attributes are present.
- No missing values or duplicates were found in the dataset.
- Features include area, bedrooms, bathrooms, parking, furnishing status, and other house amenities.

## TASK-2: Data Cleaning

- The dataset was checked for missing values.
- No missing values were found; however, a preprocessing pipeline was implemented to handle potential missing entries using median and mode imputation.
- Duplicate records were checked and removed where necessary.
- Categorical variables such as main road, guestroom, basement, air-conditioning, and furnishing status were converted into numerical format using One-Hot Encoding.
- All remaining columns were retained because they represent meaningful housing characteristics that may influence property prices.
- After preprocessing, the dataset contained only numerical features suitable for machine learning algorithms.

## TASK-3: Model Building

- The dataset was split into training and testing sets using an 80:20 ratio.
- Linear Regression was trained as the baseline model.

- Random Forest Regressor was trained as an ensemble model for comparison.
- Model Performance was evaluated using: Mean Absolute Error, Root Mean Squared,  $R^2$  Score
- Lower MAE and RMSE values indicate better prediction accuracy.
- Higher  $R^2$  values indicate better explanatory power of the model.
- Both Linear Regression and Random Forest Regressor were trained and evaluated for house price prediction. The Linear Regression model outperformed the Random Forest model, achieving a lower MAE and RMSE along with a higher  $R^2$  score of 0.6529. Therefore, Linear Regression was chosen as the final predictive model for this housing dataset.

#### **TASK-4: Visualization**

- **Histogram:** House prices are not uniformly distributed. Most properties fall within a moderate price range, with fewer high-priced properties.
- **Correlation Heatmap:** Area, bathrooms, stories, parking, and air conditioning show the strongest positive correlations with house price.
- **Scatter Plot:** The Linear Regression model predicts house prices with reasonable accuracy, as most points lie near the ideal prediction line.

This project successfully developed a house price prediction model using machine learning techniques. After data cleaning, preprocessing, visualization, and model evaluation, Linear Regression was identified as the best-performing model for this dataset. The study highlighted the key factors influencing property prices and demonstrated how data-driven insights can support more accurate real estate valuation and decision-making.